

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

Final Project Report & Weekly Work Diary

Project Title: AI Agent to Test Websites Automatically Using Natural Language

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Mentor: M. Likhita

Duration: 24th December 2025 – 20th February 2026

1. Introduction

This internship focused on building an AI-powered web testing agent capable of converting natural language instructions into automated browser test executions.

The objective of the project was to design and implement a system that:

- Accepts website URLs
- Converts natural language test instructions into structured test steps
- Executes automation using Playwright
- Performs assertions
- Generates PDF reports
- Maintains execution history
- Implements authentication and UI

The internship emphasized hands-on learning, real-world implementation, Git collaboration, and independent problem-solving.

2. Internship Timeline & Weekly Work Diary

Week 1

(29th Dec – 4th Jan)

29th Dec 2025 – Kick-off Session

- Introduction session with mentor and peers.

- Discussed internship structure and expectations.
- Understood project objective briefly.
- Daily meeting time finalized (5:30 PM – 6:30 PM).

30th Dec 2025

- Recap session.
- Discussed problem statement at high level.
- Understood that no session recordings would be provided.
- Received project documentation.

31st Dec & 1st Jan – No Classes

Assignment Given (To Complete by 5th Jan)

- Install Python.
- Learn about Virtual Environments.
- Understand why venv is used.
- Create a simple HTML login page.

Learning Outcome:

- Understood project structure.
- Set up development environment.
- Built basic HTML login form.
- Learned isolation using Python virtual environments.

Week 2

(5th Jan – 11th Jan)

5th Jan 2026

- HTML login page reviewed.
- Started working on Playwright automation.
- Implemented basic login automation script.

Subsequent Sessions

- Mentor shared handouts due to unavoidable circumstances.
- Practiced:
 - Playwright setup
 - Basic browser automation
 - Selectors
 - Click & Fill operations
 - Assertions

Learning Outcome:

- Built first working Playwright automation.
 - Understood selectors and DOM interaction.
 - Learned debugging automation failures.
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Week 3

(12th Jan – 18th Jan)

- Continued working on Playwright login demo task.
- Each student required to:
 - Demonstrate HTML login page.
 - Show Playwright automation.
 - Explain code logic and flow.

GitHub Collaboration:

- Shared GitHub usernames.
- Added as collaborators to project repository.
- Created individual branches.
- Learned:

- git clone
- git branch
- git add
- git commit
- git push

Learning Outcome:

- Understood version control workflow.
 - Practiced pushing code to individual branches.
 - Improved confidence in explaining code logic.
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Week 4**(19th Jan – 25th Jan)**

- Introduced to LangGraph basic implementation.
- Started exploring AI agent concepts.
- Received detailed handouts for independent implementation.

Tasks:

- Implement LangGraph agent basics.
- Push implementation to branch.
- Follow structured instructions strictly.

Learning Outcome:

- Understood LLM-based workflow.
 - Learned how AI agents process tasks.
 - Practiced structured code implementation from documentation.
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Week 5**(26th Jan – 1st Feb)**

- Continued LangGraph hands-on practice.
- Completed guided exercises from handouts.
- Pushed milestone-based code updates.
- Created structured project documentation folder.

Focus Areas:

- Understanding flow-based agent execution.
- Modularity in code.
- Separation of logic and execution.

Learning Outcome:

- Improved modular coding practices.
 - Learned agent-based workflow modeling.
 - Strengthened Git discipline.
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Week 6**(2nd Feb – 8th Feb)**

- Independent implementation week.
- Mentor shared detailed handouts daily.
- Required to:
 - Implement logic independently.
 - Understand code, not blindly copy.
 - Push updates regularly.

Major Progress:

- Started integrating:
 - Natural Language Processing
 - LLM API (Gemini)
 - Structured JSON generation

Learning Outcome:

- Understood LLM integration.
 - Learned handling API responses.
 - Implemented structured output parsing.
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Week 7

(9th Feb – 15th Feb)

Backend Development Phase

Implemented:

- Natural language test parser
- Playwright execution engine
- Assertion handling
- Error handling
- Optional AI-based debugging suggestions

Handled challenges:

- Async event loop errors
- Headless mode configuration
- API quota issues
- Selector failures

Learning Outcome:

- Real-world debugging experience.
- Improved system design understanding.
- Learned about rate limits and production constraints.

Week 8

(16th Feb – 20th Feb)

UI & Final Integration Phase

Implemented:

- Streamlit-based UI
- Website input field
- Natural language instruction input
- Execution logs display
- PDF report generation
- Test history tracking using SQLite
- Authentication system using bcrypt

Resolved:

- PDF download issues
- API quota exhaustion
- Selector timeout issues
- Playwright subprocess errors

Final Demo Preparation:

- Structured presentation.
- Prepared 5-minute explanation.
- Ensured working end-to-end demo.

3. Final Project Description

Project Name:

AI Agent to Test Websites Automatically Using Natural Language

Final Deliverables:

- Fully working AI-based automation platform
 - LLM-powered test step generation
 - Playwright execution engine
 - Dynamic assertion handling
 - Headless/Headed execution
 - PDF report generation
 - Test history database
 - Authentication system
 - Modular architecture
 - GitHub repository with branch structure
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4. Technologies Used

- Python
 - Playwright
 - Streamlit
 - Google Gemini API
 - LangGraph
 - SQLite
 - FPDF
 - bcrypt
 - Git & GitHub
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5. Challenges Faced

- Handling dynamic selectors
 - Debugging Playwright event loop issues
 - Managing API rate limits
 - Structuring modular architecture
 - Integrating UI with backend
 - Implementing authentication securely
 - Ensuring professional documentation
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6. Key Learnings

- Practical automation testing
 - AI integration in software systems
 - LLM-based instruction parsing
 - System architecture design
 - Modular coding practices
 - Debugging large codebases
 - Git collaboration workflow
 - Professional documentation standards
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7. Conclusion

This internship provided strong practical exposure to real-world software development workflows including AI integration, automation testing, modular architecture, debugging, and version control collaboration. The final project demonstrates the ability to design and implement an AI-powered automation framework capable of converting human instructions into executable browser tests.

The internship significantly improved my problem-solving ability, debugging skills, architectural thinking, and confidence in building full-stack AI-integrated systems.

